

Data Analysis and Storytelling

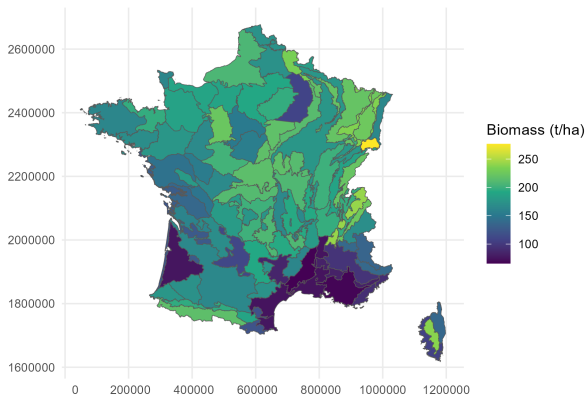
Regression analysis - continuous spatial models

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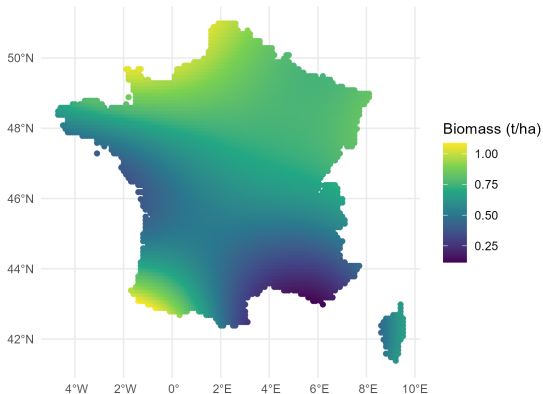
FIRS - Track 1

Two main types of spatial models



Discrete spatial units

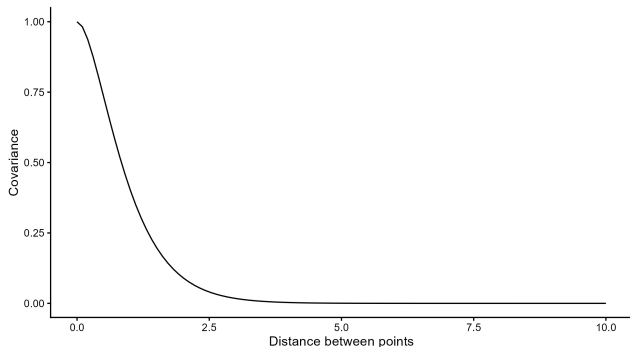
Two main types of spatial models



Continuous spatial field

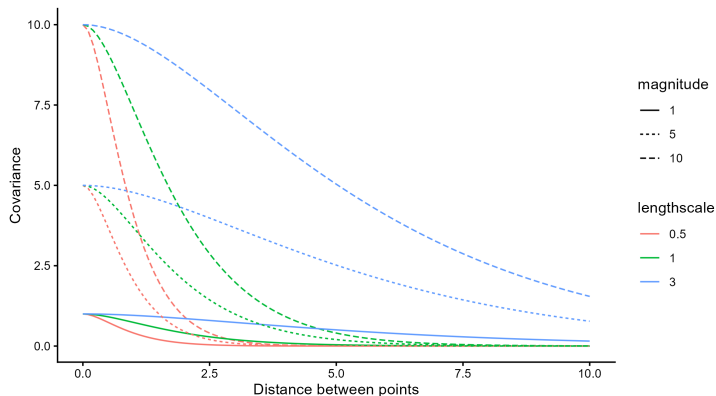
Intuition behind continuous spatial models

- Continuous spatial models rely at the core on covariance functions to fill variance-covariance matrix
- The covariance function returns a covariance between two point in space given (i) the distance between the points - closer points have larger covariation and (ii) a set of parameters to estimate

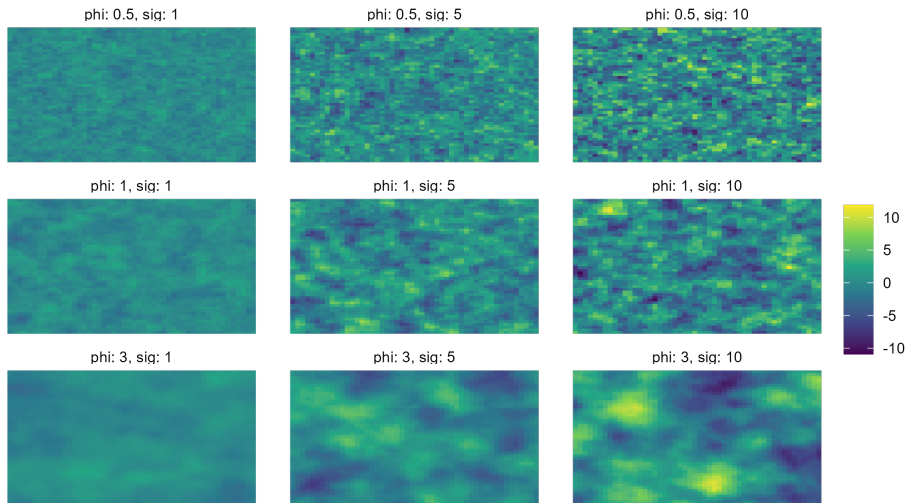


The matern covariance function

The matern covariance function is the standard choice to estimate spatial covariance. This function has two parameters: (i) lengthscale (ϕ): the speed of decay in the correlation - larger values imply smooth spatial field while smaller values imply wigglier ones, (ii) magnitude (σ): the variance in the spatial fields - large values imply more varying spatial fields.



Matern spatial fields



Covariance functions and continuous spatial models

- Relying on covariance functions to capture the spatial signal makes it possible to derive spatial fields at any spatial scale and for arbitrary locations on the fields - continuous space
- This flexibility comes with higher computing costs that scale $O(n^3)$ with the number of observations, so even for medium-sized datasets, standard laptop fail
- Sparse approaches are available to reduce the complexity and use approximate, rather than exact, solution

From covariance to spatial model

A spatially continuous regression model can therefore be written:

$$y_i \sim N(\mu_i, \sigma_{res})$$

$$\mu_i = \beta_0 + \gamma_{s_i} + \dots$$

$$(\gamma_{s_1}, \gamma_{s_2}, \dots) \sim MVN(0, \text{Matern}(\phi, \sigma_{spat}, d_{s_i, s_j}))$$

With d_{s_i, s_j} the distance between two points.

Such a model is also called gaussian process regression or kriging.

Wrap up

- Fitting spatial model with continuous spatial fields is done via a covariance function
- A natural choice is the Matern covariance function predicting covariance decay with the distance based on two parameters:
 - 1 Lengthscale: the speed of decay in the correlation with the distance
 - 2 Magnitude: the variability of the spatial field
- Such continuous models allow flexible estimation at any spatial scale and location
- However, these models can be costly to fit, sparse approach exists to circumvent this